

# Psychology 101

How Memory Works — Encoding, Storage & Forgetting

---

Sample coursework material — for product demonstration purposes.

## Chapter 1 — What Is Memory?

Memory is the mental process of encoding, storing, and later retrieving information. Far from working like a video recorder, human memory is reconstructive — each time we recall an event, we partially rebuild it, which is why memories can shift, blend, or even be distorted over time.

Psychologists generally describe memory as involving three core processes: encoding, the conversion of information into a form the brain can store; storage, the retention of that information over time; and retrieval, the process of accessing stored information when it is needed.

## Chapter 2 — The Atkinson–Shiffrin Model

One of the most influential frameworks for understanding memory is the Atkinson–Shiffrin multi-store model, proposed in 1968. It describes memory as flowing through three distinct stores, each with a different capacity and duration.

Sensory memory holds an almost exact copy of incoming sensory information for a fraction of a second — just long enough for the brain to decide what deserves attention. Information that receives attention moves into short-term memory, which holds a small amount of information for roughly 15 to 30 seconds unless it is actively rehearsed. With enough rehearsal or meaningful processing, information can move into long-term memory, a store with essentially unlimited capacity and duration.

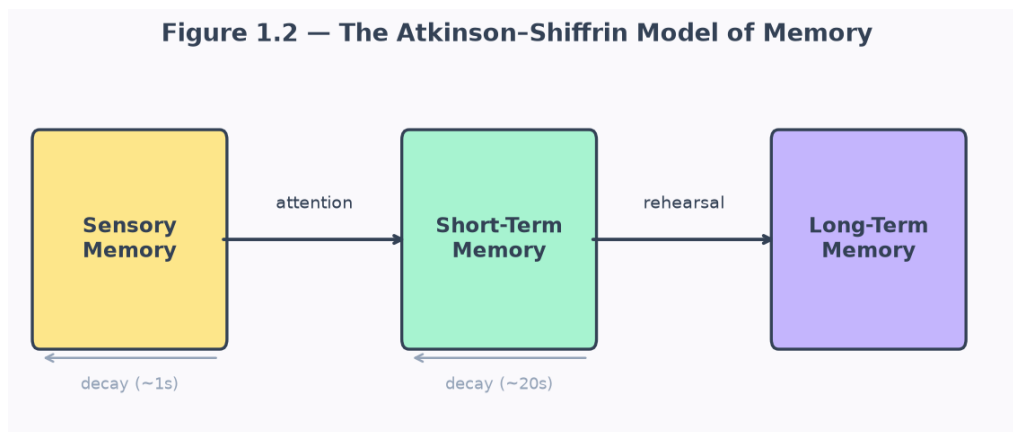


Figure 1.2 — The Atkinson–Shiffrin model: information flows from sensory memory to short-term memory to long-term memory.

## Chapter 3 — Encoding: Getting Information In

Encoding is the gateway to memory — information that is never properly encoded is never available to recall later, no matter how important it was. Psychologists distinguish several types of encoding: visual (based on how something looks), acoustic (based on how it sounds), and semantic (based on its meaning).

Semantic encoding, which connects new information to existing knowledge and personal meaning, generally produces the strongest and most durable memories — a principle known as the levels-of-processing effect. This is why relating new material to something you already understand, rather than memorizing it as an isolated fact, leads to better long-term retention.

## Chapter 4 — Short-Term and Working Memory

Short-term memory has a famously limited capacity, often cited as 'seven plus or minus two' items, though modern estimates suggest the true limit is closer to four chunks of information for most people. A chunk can be a single digit or an entire familiar word — which is why grouping information into meaningful units dramatically increases how much can be held in mind at once.

Working memory extends this idea, describing not just passive storage but the active manipulation of information — holding a phone number in mind while also dialing it, for example. Working memory is closely tied to attention and is considered a key resource underlying reasoning, comprehension, and learning.

## Chapter 5 — Long-Term Memory and Consolidation

Long-term memory is generally divided into explicit (declarative) memory — facts and events that can be consciously recalled — and implicit (procedural) memory — skills and habits, like riding a bicycle, that are expressed through behavior rather than conscious recollection.

Explicit memory is further split into episodic memory, for personally experienced events, and semantic memory, for general facts and concepts detached from any particular experience. Moving a memory from a fragile, recently formed state into a stable long-term form is called consolidation, a process strongly supported by sleep, during which the brain replays and strengthens recently formed neural connections.

## Chapter 6 — Forgetting and the Spacing Effect

Forgetting is not simply memory decay — it often reflects retrieval failure, where a memory still exists but the right cues aren't available to access it, or interference, where similar memories compete with one another. Hermann Ebbinghaus's classic research produced the 'forgetting curve,' showing that newly learned information is lost fastest in the first hours and days after learning, then more slowly afterward.

The single most well-supported way to counteract this curve is the spacing effect: review sessions spread out over increasing intervals of time produce dramatically better long-term retention than the same amount of review crammed into one sitting. This is the principle behind spaced-repetition study systems, which schedule each review just before a memory is likely to be forgotten — strengthening it at exactly the point where recall is most effortful, and therefore most effective.